



أسئلة سنوات سابقة

المادة: كالكولس 1 [فاينل ٢٠١٨ - ٢٠١٩]

Q1 Let $f(x) = x^3 - 2x + 4$, and let $f^{-1}(c^3) = c$, the value of c , is :

- A) -2 B) $-\frac{1}{2}$ C) $\frac{1}{2}$ D) 2

Q2 If the horizontal asymptote of the curve of the function $f(x) = \frac{3x^2+1}{x(Mx-2)}$ is $y = \frac{1}{3}$, then the value of M is :

- A) 6 B) 9 C) 12 D) 15

Q3 The slope of the tangent line of the curve $f(x) = (2x-5)^{50}$, at $x=2$ equals :

- A) 200 B) 100 C) -100 D) 102

Q4 Let $f(x) = \begin{cases} 2x^3+1 & \text{if } x \geq 1 \\ x^6+1 & \text{if } x < 1 \end{cases}$, then which of the following statements is true :

- A) $f(x)$ is continuous and not differentiable at $x=1$.
B) $f(x)$ is discontinuous and not differentiable at $x=1$.
C) $f(x)$ is discontinuous and differentiable at $x=1$.
D) $f(x)$ is continuous and differentiable at $x=1$.



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العادة:

Q5 If $f(x) = \sqrt{x} g(x)$, $g(25) = -40$ and $g'(25) = 2$ then the value of $f'(25)$ is :

- A) 12 B) 16 C) 4 D) 6

Q6 Let $f(x) = x^2 + ax + b$, If $f(x)$ has extreme value at $x=3$, then the value of a and b are :

- A) $a=6, b=13$ B) $a=-6, b=-13$ C) $a=6, b=-13$ D) $a=-6, b=1$

Q7 $\frac{d^{100}}{dx^{100}} \left(\int_0^x \cos(\pi t) \cdot dt \right) =$

- A) $-\pi^{99} \sin(\pi x)$ B) $\pi^{99} \sin(\pi x)$ C) $-\pi^{99} \cos(\pi x)$ D) $\pi^{99} \cos(\pi x)$

Q8 Let $f(x) = (1-x)^{\frac{1}{6}}$, then $f(x)$ increases on the interval :

- A) $(-\infty, 1]$ B) $(1, \infty)$ C) $[1, \infty)$ D) No where

Q9 Let $f(x) = (1-x)^{\frac{1}{6}}$, then $f(x)$ concave up on the interval :

- A) $(-\infty, 1]$ B) $(-\infty, 1)$ C) $(1, \infty)$ D) No where

Q10 Let $f(x) = (1-x)^{\frac{1}{6}}$, then $f(x)$ has Absolute minimum equal to :

- A) 0 B) 1 C) No absolute minimum D) 2

Q11 If $2x^3 - 3y^2 = 8$, and if $a \neq 0$ then $\frac{d^2y}{dx^2}(2a, a) =$

- A) $\frac{4a}{a} + \frac{4a^4}{a^2}$ B) $4 - 16a$ C) $4 - 8a$ D) $2 - a$



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المادة:

Q12 If $f(x) = 4x\sqrt{x+\sqrt{x}}$ then $f'(1) =$

- A) $\frac{11\sqrt{2}}{2}$ B) $4\sqrt{2} + \frac{2}{\sqrt{2}}$ C) $\frac{11\sqrt{2}}{4}$ D) $\frac{33\sqrt{2}}{2}$

Q13 Let $f(x) = \cos x$, $g(x) = a + bx + cx^2$ and let $f(0) = g(0)$, $f'(0) = g'(0)$, and $f''(0) = g''(0)$, then $\frac{d}{dx}(g \circ f)$ at $(\frac{\pi}{4})$ equals to:

- A) $-\frac{1}{2}$ B) $\frac{1}{2}$ C) $-\frac{\sqrt{3}}{4}$ D) $\frac{\sqrt{3}}{4}$

Q14 $\lim_{x \rightarrow \infty} \frac{2x\sqrt{x} + 3}{x^2 + \sqrt{x}}$

- A) 0 B) 1 C) 2 D) ∞

Q15 The length of the curve $y = \frac{4\sqrt{2}}{3}x^{\frac{1}{2}} - 1$, $0 \leq x \leq 1$

- A) $\frac{6}{12}$ B) $\frac{12}{6}$ C) $\frac{13}{6}$ D) $\frac{6}{13}$

Q16 The volume of the solid generated by revolving the region between $y = \sqrt{x}$ and $y = 0$, about the x -axis is $0 \leq x \leq 4$

- A) 18π B) 32π C) 16π D) 8π

Section 2 : written questions.

Q1 Use cylindrical shells to find the volume of cone generated when the triangle with vertices $(0,0)$, $(a,0)$, and $(a,2a)$, where $a > 0$, is revolved about the x -axis.

أسئلة سنوات سابقة

المادة: حل امتحان خايل سالكولس 1 (٢٠١٨ - ٢٠١٩)

Q1

$$f^{-1}(c^3) = c$$

$$c^3 = f(c)$$

$$c^3 = c^3 - 2c + 4$$

$$2c = 4 \rightarrow c = 2 \quad \textcircled{D}$$

Q2

$$\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{Mx^2 - 2x} = \frac{1}{3}$$

$$\lim_{x \rightarrow \infty} \frac{3x^2}{Mx^2} = \frac{1}{3}$$

$$\frac{3}{M} = \frac{1}{3} \rightarrow M = 9 \quad \textcircled{B}$$

Q3

$$f'(x) = 50(2x - 5)^{49} \cdot 2$$

$$f'(x) = 100(2x - 5)^{49}$$

$$\begin{aligned} m = f'(2) &= 100(4 - 5)^{49} \\ &= 100(-1)^{49} \\ &= -100 \quad \textcircled{C} \end{aligned}$$

Q4

cts $f(1) = 2(1)^3 + 1 = 3$

$$\lim_{x \rightarrow 1} f(x) \begin{cases} \rightarrow \lim_{x \rightarrow 1^+} 2x^3 + 1 = 3 \\ \rightarrow \lim_{x \rightarrow 1^-} x^6 + 1 = 2 \end{cases}$$

$$\lim_{x \rightarrow 1} f(x) \text{ d.n.e}$$

$\therefore f$ is not cts at $x=1$, not diff $\textcircled{B}$

Don't miss the change ... Be part of it

أسئلة سنوات سابقة

المادة:

Q5 $f'(x) = \sqrt{x} g'(x) + g(x) \cdot \frac{1}{2\sqrt{x}}$

$$f'(25) = 5 g'(25) + g(25) \cdot \frac{1}{2 \times 5}$$

$$= 5 \times 2 + -40 \cdot \frac{1}{10}$$

$$= 10 - 4 = 6 \quad \textcircled{D}$$

Q6 $(3, 4)$ extreme value

$$f(3) = 4 \quad \text{and} \quad f'(3) = 0$$

$$9 + 3a + b = 4 \quad \text{and} \quad 6 + a = 0$$

$$\rightarrow 3a + b = -5$$

$$\rightarrow a = -6$$

$$3(-6) + b = -5 \rightarrow b = 13 \quad \textcircled{D}$$

Q7 $\frac{d^{100}}{dx^{100}} \left(\int_0^x \cos(\pi t) \cdot dt \right)$

$$\rightarrow \frac{d}{dx} \int_0^x \cos(\pi t) \cdot dt = \cos(\pi x)$$

الآن نشق الجواب 99 مرة .

$$2n - 1 = 99$$

$$2n = 100 \rightarrow n = 50$$

$$\frac{d^{100}}{dx^{100}} \int_0^x \cos(\pi t) \cdot dt = (-1)^{50} \pi^{99} \cos(\pi x)$$

$$= \pi^{99} \cos(\pi x) \quad \textcircled{D}$$

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المادة:

Q8

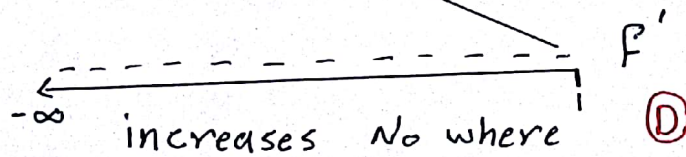
$$f(x) = (1-x)^{\frac{1}{6}}$$

* مجال الاقتران $1-x \geq 0$

$$x \leq 1$$

$$f'(x) = -\frac{1}{6} (1-x)^{-\frac{5}{6}}$$

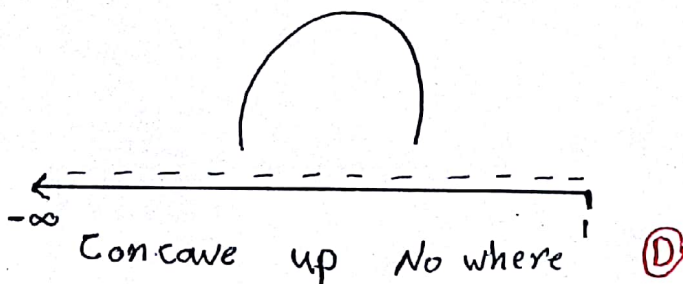
$$f'(x) = \frac{-1}{6(1-x)^{\frac{5}{6}}} \begin{cases} f'=0 & -1 \neq 0 \\ f' \text{ d.n.e} & 1-x=0 \\ & x=1 \end{cases}$$



Q9

$$f''(x) = \frac{-5}{36} (1-x)^{-\frac{11}{6}}$$

$$f''(x) = \frac{-5}{36(1-x)^{\frac{11}{6}}} \begin{cases} f''=0 & -5 \neq 0 \\ f'' \text{ d.n.e} & 1-x=0 \\ & x=1 \end{cases}$$



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المادة:

Q10

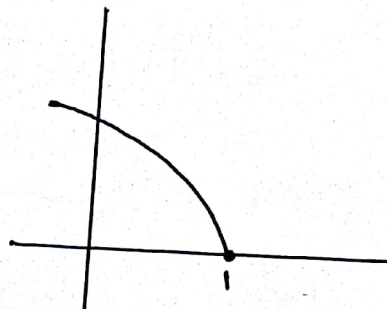
$$f(x) = \sqrt{1-x}$$

$$D_f = (-\infty, 1]$$

$$f(1) = \sqrt{0} = 0$$

at $x=1$

Abs. Min
with value 0



Q11

$$2x^3 - 3y^2 = 8$$

$$6x^2 - 6y \frac{dy}{dx} = 0 \leadsto 6x^2 = 6y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x^2}{y}$$

$$\frac{d^2y}{dx^2} = \frac{y \cdot 2x - x^2 \cdot \frac{x^2}{y}}{y^2}$$

$$\frac{d^2y}{dx^2} (2a, a) = \frac{a \cdot 2 \cdot 2a - 4a^2 \cdot \frac{4a^2}{a}}{a^2}$$

$$= \frac{4a^2 - 16a^3}{a^2} = 4 - 16a \quad \textcircled{B}$$

Q12

$$f'(x) = 4x \cdot \frac{1 + \frac{1}{2\sqrt{x}}}{2\sqrt{x+\sqrt{x}}} + \sqrt{x+\sqrt{x}} \cdot 4 = 4 \cdot \frac{3}{4\sqrt{2}} + 4\sqrt{2}$$

$$= \frac{3}{\sqrt{2}} + 4\sqrt{2}$$

$$f'(1) = 4 \cdot \frac{1 + \frac{1}{2}}{2\sqrt{2}} + 4\sqrt{2}$$

$$= \frac{11}{\sqrt{2}} = \frac{11\sqrt{2}}{2} \quad \textcircled{A}$$

$$= 4 \cdot \frac{3/2}{2\sqrt{2}} + 4\sqrt{2}$$

Don't miss the change ... Be part of it



أسئلة سنوات سابقة

العادة:

Q13 $\frac{d}{dx} (g(f(x))) = g'(f(x)) \cdot f'(x)$

$$\begin{aligned}\left. \frac{d}{dx} (g(f(x))) \right|_{x=\frac{\pi}{4}} &= g'(f(\frac{\pi}{4})) \cdot f'(\frac{\pi}{4}) \\ &= g'(\cos \frac{\pi}{4}) \cdot (-\sin \frac{\pi}{4}) \\ &= g'(\frac{1}{\sqrt{2}}) \cdot (-\frac{1}{\sqrt{2}}) \\ &= \frac{-1}{\sqrt{2}} \cdot \frac{-1}{\sqrt{2}} = \frac{1}{2}\end{aligned}$$

$$f(x) = \cos x$$

$$g(x) = a + bx + cx^2$$

$$f'(x) = -\sin x$$

$$g'(x) = b + 2cx$$

$$f''(x) = -\cos x$$

$$g''(x) = 2c$$

$$\begin{array}{c|c|c} f(0) = g(0) & f'(0) = g'(0) & f''(0) = g''(0) \\ 1 = a & 0 = b & -1 = 2c \\ & & c = -\frac{1}{2} \end{array}$$

$$\therefore g'(x) = -x \quad \textcircled{B}$$

Q14 $\lim_{x \rightarrow \infty} \frac{2x\sqrt{x} + 3}{x^2 + \sqrt{x}}$

$$\begin{aligned}\rightarrow \lim_{x \rightarrow \infty} \frac{2x\sqrt{x}}{x^2} &= \lim_{x \rightarrow \infty} \frac{2x^{3/2}}{x} \\ &= \lim_{x \rightarrow \infty} \frac{2}{x^{1/2}} = \frac{2}{\infty} = 0 \quad \textcircled{A}\end{aligned}$$



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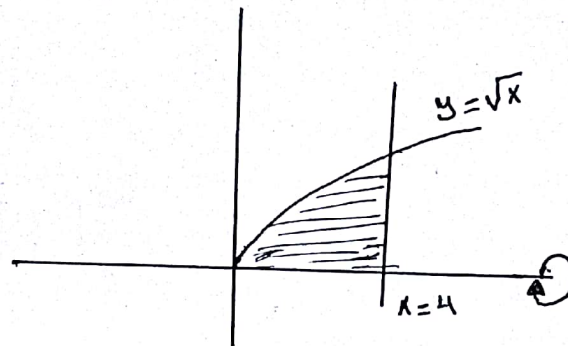
المادة:

Q15 $\frac{dy}{dx} = 2\sqrt{x}^{1/2}$

$$\left[\frac{dy}{dx}\right]^2 = 8x$$

$$\begin{aligned} L &= \int_0^1 \sqrt{1+8x} \cdot dx = \int_0^1 (1+8x)^{1/2} \cdot dx \\ &= \left[\frac{(1+8x)^{3/2}}{\frac{3}{2} \cdot 8} \right]_0^1 \\ &= \frac{27}{12} - \frac{1}{12} = \frac{13}{6} \quad \text{C} \end{aligned}$$

Q16



$$\begin{aligned} V &= \pi \int_0^4 (\sqrt{x})^2 - 0^2 \cdot dx \\ &= \pi \int_0^4 x \cdot dx \\ &= 8\pi \quad \text{D} \end{aligned}$$



أسئلة سنوات سابقة

المادة:

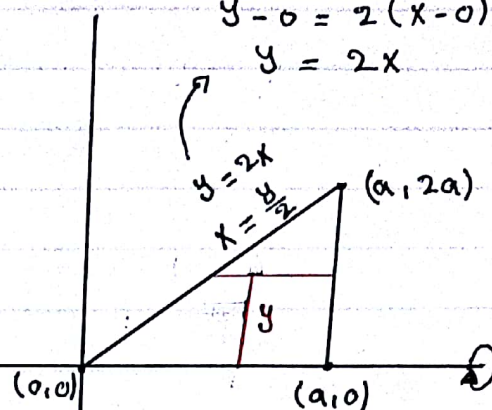
Section 2:

Q1

$$m = \frac{2a-0}{a-0} = 2$$

$$y-0 = 2(x-0)$$

$$y = 2x$$



* عند دورانها على محور x ينشأ شكل مخروطي.

$$V = 2\pi \int_a^{2a} y \left(a - \frac{y}{2}\right) dy$$

$$= 2\pi \int_a^{2a} \left(ay - \frac{y^2}{2}\right) dy$$

$$= 2\pi \left(a \cdot \frac{y^2}{2} - \frac{y^3}{6} \right) \Big|_0^{2a}$$

$$= 2\pi \left(a \cdot \frac{4a^2}{2} - \frac{8a^3}{6} \right)$$

$$= 2\pi \left(2a^3 - \frac{4}{3}a^3 \right)$$

$$= 4a^3\pi - \frac{8}{3}a^3\pi = \frac{4a^3\pi}{3}$$



سئلة سنوآت سابقة

مادة: فائناال كالكولاس ١

Q1:- If $y = \sin^{-1} \sqrt{x+3}$, then $\frac{dy}{dx} =$

a) $\frac{1}{2\sqrt{x+3}\sqrt{-2+x}}$

c) $\frac{1}{2\sqrt{x+3}\sqrt{-2-x}}$

Q2:- If $f(x) = \sqrt{\frac{x+2}{x-2}}$, $x > 2$, then $f^{-1}(x) =$

a) $\frac{2x^2+2}{x^2-1}$

c) $\frac{x^2+1}{x^2-1}$

Q3:- If $\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - 3}{x-3} =$

a) $\frac{1}{3}$

c) $\frac{1}{6}$

Q4:- $\lim_{x \rightarrow 0} \frac{x - \sin x}{x \cos x} =$

a) 0

c) 1

b) 2

d) -1

Q5:- The domain of the function $f(x) = \frac{1}{\sqrt{4-x^2}}$ is :-

a) $(-2, 2)$

c) $(-2, -\infty)$

b) $(-\infty, -2) \cup (2, \infty)$

d) $[-2, 2]$

Q6:- $y = (\cosh x)^{\cosh x}$, then $\frac{dy}{dx} =$

a) $(\cosh x)^{\cosh x} [1 + \ln(\cosh x) \sinh x]$

b) $(\cosh x)^{\cosh x} [1 + \ln(\cosh x)] \cosh x$

c) $(\cosh x) [1 + \ln(\cosh x)] \sinh x$

d) $(\cosh x)^{\cosh x} [1 + \ln(\sinh x)] \sinh x$

Q7:- The range of the function $f(x) = 1 - \sqrt{x-2}$ is :-

a) $(-\infty, 2]$

c) $[1, \infty)$

b) $(-\infty, 1]$

d) $[2, \infty)$

Q8:- If $\ln(\ln(x-2)) = 0$, then =

a) $e^2 + 2$

c) $2e + 2$

b) $e + 2$

d) $2e + 1$



مسئلة سنوات سابقة

المادة:

Q9:- If $f(x) = x|x|$, then f has an inflection point at

$x =$

a) -2

b) 2

c) 1

d) 0

Q10:- The area of the region between the curves $y = x$ and $y = \sqrt{x}$ is :-

a) $\frac{1}{2}$

b) 1

c) $\frac{1}{3}$

d) $\frac{1}{6}$

Q11:- $\lim_{x \rightarrow 0^+} (x+1)^{\frac{2}{x}}$:-

a) e

b) e^2

c) $e^{\frac{1}{2}}$

d) e^3

Q12:- $\int \cosh(3-2x) dx =$

a) $\frac{1}{2} \sinh(3-2x) + C$

b) $-\frac{1}{2} \sinh(3-2x) + C$

c) $-2 \sinh(3-2x) + C$

d) $\sinh(3-2x) + C$

Q13:- $\int_1^2 \frac{x+1}{x} dx =$

a) $\ln 2$

b) 2

c) $1 + \ln 2$

d) $2 + \ln 2$

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مسئلة سنوات سابقة

المادة:

Q14:- $\int \frac{4 \tan^{-1}(x)}{1+x^2} dx$

a) $\ln[1+x^2] + C$

b) $2 \ln[1+x^2] + C$

c) $2(\tan^{-1}(x))^2 + C$

d) $\tan^{-1}x + C$

Q15:- The horizontal asymptote of $f(x) = \frac{\sqrt{x^2+1}}{2x-1}$ are:-

a) $x = \pm \frac{1}{2}$

b) $x = \pm 1$

c) $x = \pm \frac{1}{3}$

d) $x = \pm \frac{1}{4}$

Q16:- let $f(x) = \ln(x^2-4)$, then f continuous on.

a) $[-2, 2]$

b) $(-\infty, -2) \cup (2, \infty)$

c) $(-2, 2)$

d) $(-\infty, 2) \cup (2, \infty)$

Q17:- Suppose f is a differentiable function such that $f(g(x)) = 3x$.

a) $\frac{1}{1+x^2}$

b) $\frac{2}{1+4x^2}$

c) $\frac{3}{1+9x^2}$

d) $\frac{1}{1+16x^2}$



سئلة سنوات سابقة

عادة:

part(B) :- If $f(x) = x^3 + x^2 + 1$, then

1) the decreasing intervals of f are

2) f has arelative maximum at $x = \dots\dots\dots$

3) The concave down interval of f is

1) the decreasing intervals of f are

2) f has a relative maximum at $x = \dots\dots\dots$

3) The concave down interval of f is

Solution:-

Q1:- $y = \sin^{-1}(\sqrt{x+3}) \Rightarrow \frac{dy}{dx} = ?!$

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x+3}(\sqrt{x+3})^2}$$

$$= \frac{1}{2\sqrt{x+3}\sqrt{1-x-3}}$$

$$= \frac{1}{2\sqrt{x+3}\sqrt{-2-x}}$$

--- (C)

Q2:- $f(x) = \sqrt{\frac{x+2}{x-2}}$, $x > 2$, $f^{-1}(x) =$

$$y = \sqrt{\frac{x+2}{x-2}}$$

$$y^2 = \frac{x+2}{x-2} \longrightarrow x^2 = \frac{y+2}{y-2} \longrightarrow x^2 y - 2x^2 = y + 2$$

$$x^2 y - y = 2x^2 + 2$$

$$y(x^2 - 1) = 2x^2 + 2$$

$$y = \frac{2x^2 + 2}{x^2 - 1}$$

(a) ---

Don't miss the change ... Be part of it

Don't miss the change ... Be part of it



مسئلة سنوات سابقة

المادة:

Q3:- $\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - 3}{x-3}$

$$\lim_{x \rightarrow 3} \frac{\sqrt{x+6} - 3}{(x-3)} \cdot \frac{(\sqrt{x+6} + 3)}{(\sqrt{x+6} + 3)}$$

$$= \lim_{x \rightarrow 3} \frac{x+6-9}{(x-3)(\sqrt{x+6} + 3)}$$

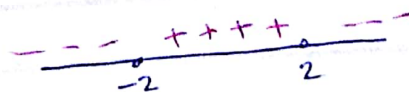
$$= \lim_{x \rightarrow 3} \frac{(x-3)}{(x-3)(\sqrt{x+6} + 3)} = \frac{1}{6} \text{ --- (C)}$$

Q4:- $\lim_{x \rightarrow 0} \frac{x - \sin x}{x \cos x} = \lim_{x \rightarrow 0} \frac{\sin x}{x \cos x}$

$$1 = \lim_{x \rightarrow 0} \frac{\tan x}{x} \rightarrow 1 - 1 = 0 \text{ --- (a)}$$

Q5:- $\frac{1}{\sqrt{4-x^2}}$

$$4 - x^2 \geq 0$$
$$(2-x)(2+x)$$



$$(-2, 2) \text{ --- (a)}$$

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مسئلة سنوات سابقة

المادة:

Q6:- $y = (\cosh x)^{\cosh x}$

$$y = \cosh x \ln \cosh x.$$

$$\frac{1}{y} \frac{dy}{dx} = \cosh x \frac{\sinh x}{\cosh x} + \ln \cosh x \sinh x.$$

$$\frac{1}{y} \frac{dy}{dx} = \sinh x + \ln \cosh x \sinh x.$$

$$\frac{1}{y} \frac{dy}{dx} = (1 + \ln \cosh x) \sinh x.$$

$$\frac{dy}{dx} = y (1 + \ln \cosh x) \sinh x.$$

$$\frac{dy}{dx} = (\cosh x)^{\cosh x} (1 + \ln \cosh x) \sinh x \quad \sim (a)$$

Q7:- $f(x) = 1 - \sqrt{x-2}$
 $\sqrt{x-2} \geq 0$
 $1 - \sqrt{x-2} \leq 1$

$$1 - \sqrt{x-2} \leq 1, \quad (-\infty, 1] \quad \sim (b)$$

Q8:- $\ln(\ln(x-2)) = 0$

$$\ln(x-2) = 1$$

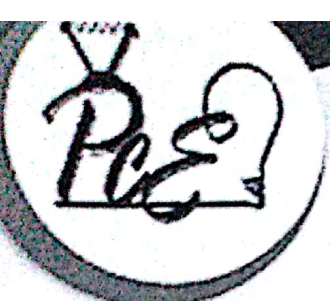
$$x-2 = e$$

$$x = e+2 \quad \sim (b)$$

Q9:- $x|x| \rightarrow \begin{cases} x^2, & x \geq 0 \\ -x^2, & x < 0 \end{cases} \rightarrow f(x) = \begin{cases} 2x, & x \geq 0 \\ -2x, & x < 0 \end{cases} \quad \sim (d)$

$$f''(x) = \begin{cases} 2, & x \geq 0 \\ -2, & x < 0 \end{cases}$$

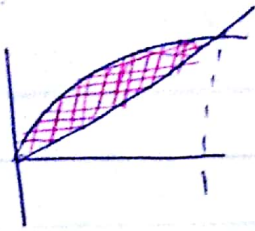
Don't miss the change ... Be part of it



مسئلة سنوات سابقة

المادة:

Q10:-



$$\int_0^1 (\sqrt{x} - x) dx$$

$$\frac{2}{3} x^{3/2} - \frac{x^2}{2} \rightarrow \frac{2 \cdot \frac{2}{3}}{2 \cdot 3} - \frac{1 \cdot 3}{2 \cdot 3} = \left[\frac{1}{6} \right] \quad \text{--- (d)}$$

Q11:- $\lim_{x \rightarrow 0^+} (x+1)^{\frac{2}{x}} = e^2 \quad \text{--- (b)}$

Q12:- $\int \cosh(3-2x) \cdot dx$

$$= -\frac{1}{2} \sinh(3-2x) + C \quad \text{--- (b)}$$

Q13:- $\int_1^2 \frac{x+1}{x} \cdot dx$

$$\int_1^2 \frac{x}{x} \cdot dx + \int_1^2 \frac{1}{x} \cdot dx \rightarrow x \Big|_1^2 + \ln x \Big|_1^2$$

$$= 1 + \ln 2 - \ln 1$$

$$1 + \ln 2 \quad \text{--- (c)}$$

Q14:- $\int \frac{4 \tan^{-1}(x)}{1+x^2} \cdot dx$

$$z = \tan^{-1} x \quad ; \quad \frac{dz}{dx} = \frac{1}{1+x^2} \quad = \int 4z dz = \frac{4z^2}{2} = (2 \tan^{-1}(x))^2 + C$$

$$\text{--- (c)}$$



مسئلة سنوات سابقة

المادة:

Q15:- $\lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1}}{2x-1}$

$\lim_{x \rightarrow \infty} \frac{x}{2x} \rightarrow y = \pm \frac{1}{2} \sim \textcircled{a}$

Q16:-

$f(x) = \ln(x^2-4)$

$x^2-4 > 0$

$(x-2)(x+2) = 0$

$(-\infty, -2) \cup (2, \infty)$



Q17:-

$f(g(x)) = 3x$

$f'(x) = 1 + (f(x))^2$

$f'(3x) = \frac{3}{1+9x^2} \sim \textcircled{c}$

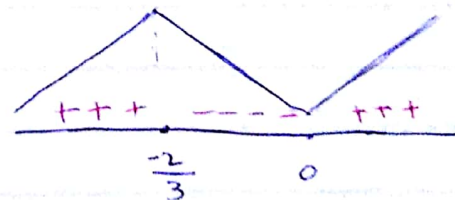
[B] $f(x) = x^3 + x^2 + 1$

$f'(x) = 3x^2 + 2x$

$3x^2 + 2x = 0$

$x(3x+2) = 0$

$\boxed{x \leq 0} \rightarrow x = -\frac{2}{3}$

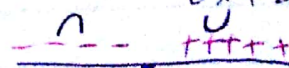


Q $[-\frac{2}{3}, 0]$

2) relative maximum at $x = -\frac{2}{3}$

3) $f''(x) = 6x + 2$

$6x + 2 = 0; \boxed{x = -\frac{1}{3}}$
 $\boxed{x = 0}$



Don't miss the change ... Be part of it $\# (-\infty, -\frac{1}{3})$